The Journal of Open Source Software

PhaseFieldPet: An Open-Source Phase-Field Modeling Software for Heterogeneous Architectures

Shore Salle Chota [1,4¶], **Martin Reder** [1,2], **Daniel Schneider** [1,2], **Harald Koestler** [4], and **Britta Nestler** [1,2,3]

1 Institute for Applied Materials (IAM), Karlsruhe Institute of Technology, Karlsruhe, 76131, Germany **2** Institute of Digital Materials Science (IDM), Karlsruhe University of Applied Sciences, Karlsruhe, 76133, Germany **3** Institute of Nanotechnology (INT), Karlsruhe Institute of Technology, Karlsruhe, 76131, Germany **4** Chair for System Simulation, Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 14 March 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Phase field method has emerged as a powerful computational tool for simulating complex, evolving interfaces at mesoscale in materials science and engineering, fluid dynamics, cell migration and other fields. However, achieving scalability and efficiency for parallel simulations of multicomponent multiphase systems across diverse hardware architectures remains a challenge. This paper presents PhaseFieldPet, an open-source, message passing interface (mpi) based software package for large-scale phase field simulations, specifically designed to leverage heterogeneous architectures such as CPUs and GPUs. It is built upon the Portable, Extensible Toolkit for Scientific Computation (PETSc). The software's modular design facilitates and easily integrates various phase field models such as Multiphase-Field and Multi-Order Parameter models with various choice of gradient and potential energy contributions.

Phase Field Equation

The phase-field approach represents surfaces and interfaces implicitly using continuous scalar fields (order parameter) $\phi_\alpha(\mathbf{r}, t)$, $\alpha \in \{1, 2, \dots, N\}$, which are constant in bulk phases and with smooth transition across the interface. The order parameters $\phi_\alpha(\mathbf{r}, t)$ represent different grains or phases such as solid, liquid, gas and their state like crystallographic orientation, polarization, volume fraction. For example, in a solid-solid phase transition, a solid phase can be represented by N order parameters $\phi_\alpha(\mathbf{r}, t)$ based on N crystallographic orientations or grains.

Microstructure evolution, and hence evolution of order parameter $\phi_\alpha(\mathbf{r}, t)$ can be obtained from functionals of entropy, or free energy, or grand potential (Hötzer et al., 2018). Following an energy approach, one can write the total free energy functional of the system as

$$\mathcal{F}(\phi, \nabla\phi, \dots) = \int_v f dv = \int_v f_{\text{grad}}(\phi, \nabla\phi) + f_{\text{pot}}(\phi) + f_{\text{bulk}}(\phi, \dots)\, dv.$$

There exists various formulations to $f_{\text{grad}}(\phi, \nabla\phi)$ and $f_{\text{pot}}(\phi)$ by many scholars in the phase field community (Daubner et al., 2023). As an example, Nestler et al. (2005) formulate these terms as

$$f_{\text{grad}}(\phi, \nabla\phi) = \varepsilon \sum_\alpha \sum_{\beta>\alpha} \gamma_{\alpha\beta} |\phi_\alpha \nabla\phi_\beta - \phi_\beta \nabla\phi_\alpha|^2,$$

$$f_{\text{pot}}(\phi) = \frac{16}{\varepsilon\pi^2}\sum_{\alpha}\sum_{\beta>\alpha}\gamma_{\alpha\beta}\phi_\alpha\phi_\beta + \frac{1}{\varepsilon}\sum_{\alpha}\sum_{\beta>\alpha}\sum_{\delta>\beta}\gamma_{\alpha\beta\delta}\phi_\alpha\phi_\beta\phi_\gamma.$$

See Daubner et al. (2023) table B.4 for other formulations of these terms and their explanations currently considered in PhaseFieldPet. The bulk contribution $f_{\text{bulk}}(\phi, ...)$ can be of various type depending on the problem at hand such as chemical, thermal, mechanical, electrical, magnetic and etc.

Following total energy minimization principle of the system, the phase field evolution equations for each $\phi_\alpha(\mathbf{r}, t)$ are in general derived by variational derivatives of the functional $\mathcal{F}(\phi, \nabla\phi, ...)$ and are given by Allen-Cahn or time-dependent Ginzburg–Landau equations for each order parameter by

$$\frac{\partial\phi_\alpha}{\partial t} = -L\frac{\delta\mathcal{F}}{\delta\phi_\alpha} = -L\left(\frac{\partial f}{\partial\phi_\alpha} - \nabla\cdot\frac{\partial f}{\partial\nabla\phi_\alpha}\right),$$

where L is kinetic coefficient. The resulting set of nonlinear parabolic differential equations define the multi-order parameter (MOP) phase-field model and are selected in PhaseFieldPet via option `pfe_mop`.

Multiphase-field models restrict the phase fields such that $\phi_\alpha(\mathbf{r}, t) \in [0, 1]$, $\sum_\alpha \phi_\alpha = 1$. Nestler et al. (2005) introduced a Lagrange multiplier λ to restrict the summation, yielding Allen-Cahn type Phase field equation

$$\frac{\partial\phi_\alpha}{\partial t} = -L\left(\frac{\partial f}{\partial\phi_\alpha} - \nabla\cdot\frac{\partial f}{\partial\nabla\phi_\alpha}\right) + \lambda.$$

This is Lagrangian based Multiphase-field model (`mpfl`), and is chosen in PhaseFieldPet by `pfe_mpfl`. Steinbach & Pezzolla (1999) rewrote the phasefield evolution equations by the sum of binary interactions

$$\frac{\partial\phi_\alpha}{\partial t} = -\frac{1}{\tilde{N}\epsilon}\sum_{\beta\neq\alpha}^{\tilde{N}} M_{\alpha\beta}\left(\frac{\delta\mathcal{F}}{\delta\phi_\alpha} - \frac{\delta\mathcal{F}}{\delta\phi_\beta}\right),$$

where $M_{\alpha\beta}$ is a mobility matrix. This Multiphase-field model (`mpf`), is chosen in PhaseFieldPet via `pfe_mpf`.

We refer interested readers to Daubner et al. (2023), Moelans et al. (2008) and chapter seven of the book by Provatas & Elder (2010) for a detailed overview of various phase-field formulations and associated evolution equations.

Statement of need

For the past couple of decades, phase-field software has been being developed and used with in house codes, and open source phase field software started to be available from 2007 (Hong & Viswanathan, 2020). Many existing open source software are limited to one or two spatial dimensions, focus on binary systems, use only one type time step solver (usually explicit time stepping), work only on one CPU core (serial code) or are not capable of using heterogeneous compute resources available such as GPUs for compute and energy efficiency. Notable large scale, distributed computing capable open source phase-field software that mainly target CPUs include: MOOSE (Schwen et al., 2017), PRISMS-PF (DeWitt et al., 2020), OpenPhase (Tegeler et al., 2017), MicroSim (Dutta et al., 2025). Among proprietary, distributed machines capable software is PACE3D (Hötzer et al., 2018). Table 1 below gives a comparison of selected

DRAFT

state of the art software for Allen-Cahn (and variations thereof) type phase-field model solvers with online tutorial available, able to run on distributed - large scale hardware architectures.

Software	Various Time Step Solver	3D capability	GPU capability	Phase Field Model	Spatial Discretization	Remark
PRISMS-PF	No	Yes	No	mop	FEM	Built on Deal.II
MicroSim	No	Yes*	Yes*	mop, mpfl*	FDM	* mpfl is not 3D and GPU capable yet
OpenPhase	No	Yes	No	mpf	FDM	GPU capability ongoing
PACE3D	No	Yes	No	mpfl, mpf	FDM	Proprietary
MOOSE	Yes	Yes	Yes**	mop, mpfl	FEM	**Possible for Nonlinear solver (via PETSc)
PhaseFieldPet	Yes	Yes	Yes	mop, mpfl, mpf	FDM	FEM possible

Table 1: Actively developed MPI-capable phase-field software.

PhaseFieldPet is a finite difference method (FDM) based software built on top of Time Step (TS) solver from PETSc (Abhyankar et al., 2018). It is based on the previous work (Daubner et al., 2023) including all the different model formulations compared therein. PhaseFieldPet extends the numerical solutions to 3D, an arbitrary amount of N phases and includes various bulk driving forces according to Hoffrogge et al. (2025). It fills the aforementioned gaps in existing software by combining the following features:

1. Enabling of multiphase simulations in 1D / 2D / 3D.
2. Decoupling the numerical solution methods from the physical modeling such that one can choose various solution methods without restricting to one time step solver (i.e. one can use methods like semi implicit, implicit time stepping algorithms, various underlying nonlinear solver, linear solvers and preconditioners, etc) based on composability features of PETSc (Balay et al., 2024).
3. Executable on single core, multicore to multi node High Performance Computing clusters/supercomputers coupled with accelerators such as GPUs (Mills et al., 2021).
4. Flexibility to easily switch between various phase-field models and energy contributions at run time.

Usage

To use PhaseFieldPet, PETSc (Balay et al., 2024) needs to be installed, compiled using
`make PhaseFieldPet`,
run the executable generated with `mpiexec/mpirun/srun` or similar other tools. For example using mpiexec with 4 mpi processes:
`mpiexec -n 4 PhaseFieldPet [options]`,
where options can be various gradient, potential, bulk energy contributions or numerical methods used to solve. For details of usage and examples, including your own energy contributions, see the associated GitHub to this paper.

Example Result

Figure 1 shows a PhaseFieldPet result of static triple junctions on 256 x 256 x 3 grids using adaptive Backward Euler (fully implicit) time step solver and `-grad_dot -pot_toth -pfe_mpfl` options. See GitHub for scalability results.

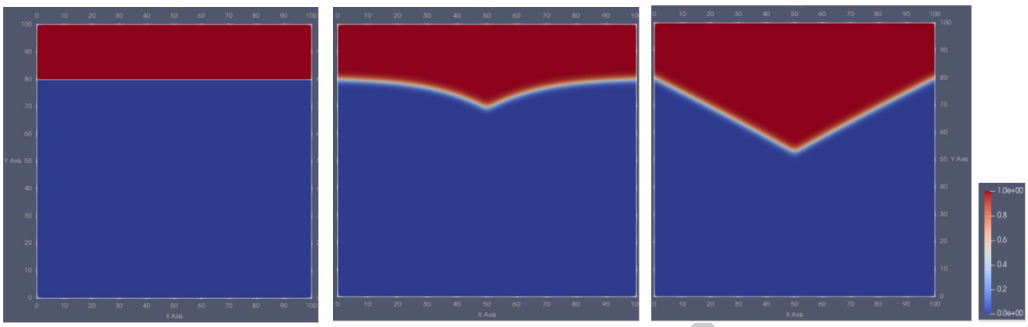

Figure 1: Time evolution of $\phi_1(\mathbf{r}, t)$: Left: Initial, Middle: Intermediate, Right: Final state.

Conclusions

PhaseFieldPet provides users with flexible methods to solve multiphase-field equations, along with various energy contributions on heterogeneous hardware architectures including CPUs and GPUs. More specifically, a user can include specific gradient, potential, bulk driving energy terms and choose the type of the phase field equation, and type of numerical algorithm to use in order to solve the differential equation. Inline with PACE3D software (Hötzer et al., 2018) and its extension, the future version of PhaseFieldPet will include various other modules corresponding to different applications of general multiphysics multiphase-field methods.

Acknowledgement

We acknowledge discussions with Dr. Simon Daubner during the genesis of this project. The authors are grateful for the scientific support and HPC resources provided by the Erlangen National High Performance Computing Center (NHR@FAU) of the Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU). The hardware is funded by the German Research Foundation (DFG).

Funding

The author SSC thanks the NHR-Verein e.V for supporting this work/project within the NHR Graduate School of National High Performance Computing (NHR). Furthermore, DS and BN acknowledge support from the Helmholtz association within programme MTET, no.38.04.04.

References

Abhyankar, S., Brown, J., Constantinescu, E. M., Debojyoti Ghosh, Smith, B. F., & Zhang, H. (2018). PETSc/TS: A modern scalable ODE/DAE solver library. In *arXiv e-preprints*. https://arxiv.org/abs/1806.01437

Balay, S., Abhyankar, S., Adams, M. F., Benson, S., Brown, J., Brune, P., Buschelman, K., Constantinescu, E. M., Dalcin, L., Dener, A., Eijkhout, V., Faibussowitsch, J., Gropp, W. D., Hapla, V., Isaac, T., Jolivet, P., Karpeev, D., Kaushik, D., Knepley, M. G., … Zhang, J. (2024). *PETSc Web page*. https://petsc.org/

Daubner, S., Hoffrogge, P. W., Minar, M., & Nestler, B. (2023). Triple junction benchmark for multiphase-field and multi-order parameter models. *Computational Materials Science*, *219*, 111995. https://doi.org/10.1016/j.commatsci.2022.111995

DeWitt, S., Rudraraju, S., Montiel, D., Andrews, W. B., & Thornton, K. (2020). PRISMS-PF: A general framework for phase-field modeling with a matrix-free finite element method. *Npj Computational Materials*, *9*, 29. https://doi.org/10.1038/s41524-020-0298-5

DRAFT

Dutta, T., Mohan, D., Shenoy, S., Attar, N., Kalokhe, A., Sagar, A., Bhure, S., Swaroop S. Pradhan, SS., Praharaj, J., Mridha, S., Kushwaha, A., Shah, V., Gururajan, M. P., Venkatesh Shenoi, V., Phanikumar, G., Bhattacharyya, S., & Choudhury, A. (2025). MicroSim: A high-performance phase-field solver based on CPU and GPU implementations. *Computational Materials Science*, *246*, 113438. https://doi.org/10.1016/j.commatsci.2024.113438

Hoffrogge, PW., Daubner, S., Schneider, D., Nestler, B., Zhou, B., & Eiken, J. (2025). Triple junction benchmark for multiphase-field models combining capillary and bulk driving forces. *Modelling and Simulation in Materials Science and Engineering*, *33*, 015001. https://doi.org/10.1088/1361-651X/ad8d6f

Hong, Z., & Viswanathan, V. (2020). Open-Sourcing Phase-Field Simulations for Accelerating Energy Materials Design and Optimization. *ACS Energy Letters*, *10*, 3254–3259. https://doi.org/10.1021/acsenergylett.0c01904

Hötzer, J., Reiter, A., Hierl, H., Steinmetz, P., Selzer, M., & Nestler, B. (2018). The parallel multi-physics phase-field framework Pace3D. *Journal of Computational Science*, *26*, 1–12. https://doi.org/10.1016/j.jocs.2018.02.011

Mills, R. T., Adams, M. F., Balay, S., Brown, J., & Dener, A. (2021). Toward performance-portable PETSc for GPU-based exascale systems. *Parallel Computing*, *108*, 102831. https://doi.org/10.1016/j.parco.2021.102831

Moelans, N., Blanpain, B., & Wollants, P. (2008). An introduction to phase-field modeling of microstructure evolution. *Calphad*, *134*, 268–294. https://doi.org/10.1016/j.calphad.2007.11.003

Nestler, B., Garcke, H., & Stinner, B. (2005). Multicomponent alloy solidification: Phase-field modeling and simulations. *Physical Review E*, *71*, 041609. https://doi.org/10.1103/PhysRevE.71.041609

Provatas, N., & Elder, K. (2010). *Phase-field methods in materials science and engineering*. Wiley-VCH Verlag GmbH & Co. KGa. https://doi.org/10.1002/9783527631520

Schwen, D., Aagesen, L. K., Peterson, J. W., & Tonks, M. R. (2017). Rapid multiphase-field model development using a modular free energy based approach with automatic differentiation in MOOSE/MARMOT. *Computational Materials Science*, *132*, 36–45. https://doi.org/10.1016/j.commatsci.2017.02.017

Steinbach, I., & Pezzolla, F. (1999). Generalized field method for multiphase transformations using interface fields. *Physica D*, *134*, 385–393. https://doi.org/10.1016/S0167-2789(99)00129-3

Tegeler, M., Shchyglo, O., Kamachali, R. D., Monas, A., Steinbach, I., & Sutmann, G. (2017). Parallel multiphase field simulations with OpenPhase. *Computer Physics Communications*, *215*, 173–187. https://doi.org/10.1016/j.cpc.2017.01.023

DRAFT